

Lecture 10 - Regression Discontinuity

Alice Wu

UW Madison Econ 695

- RD Framework
 - Sharp Identification
 - Fuzzy Identification
- Empirical Examples:
 - Medicare (running variable = age)
 - Access to student loan and college enrollment in Chile (running variable = test score)
 - Economics major and earnings from UCSC (running variable = GPA)

- Interested in “causal effect” of x on y , holding other things constant.
- Gold standard: randomized experiment (RCT)
- **Non-experimental methods:**
 - 1 OLS and reweighting counterfactuals (Oaxaca; DFL)
 - 2 IV
 - 3 Diff in diff
 - 4 **Regression Discontinuity**

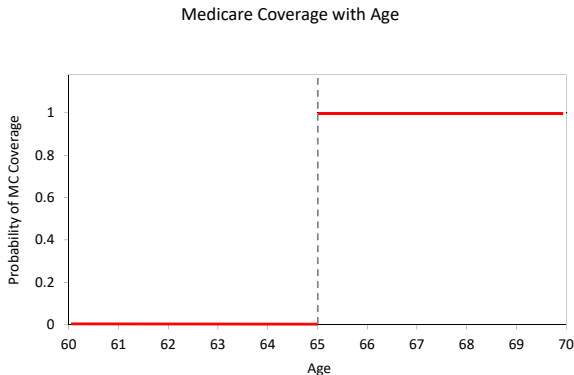
Regression Discontinuity: General Ideas

- RD arises when assignment (D_i) is based on a **discontinuous function** of some “random” or “as good as random” *running variable* x_i . For example, a sharp RD takes the form:

$$D_i = \begin{cases} 0 & x < c \\ 1 & x \geq c \end{cases}$$

- Units on either side of the discontinuity are “the same”.* The distribution of r.v. and other observable characteristics should be smooth around the cutoff.
- Examples:
 - election outcomes (rv=vote share). Let D_i indicate being elected, $D_i = 1[x_i \geq 0.50]$ based on majority rule.
 - admission or financial aid based on test scores (rv=score)
 - rules based on job tenure, duration of marriage etc (rv=duration)
 - rules based on age (rv=age). Let D_i indicate eligibility for Medicare, $D_i = 1[x_i \geq 65]$ for rv $x = \text{age}$.

Regression Discontinuity: General Ideas

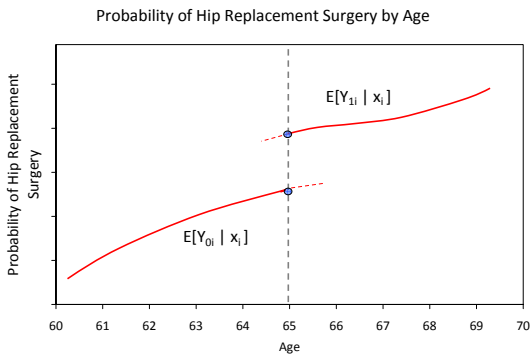


We could exploit this jump in insurance coverage to look at changes in health care usage, and health outcomes...

- x_i = running variable
- D_i = assignment status: $D_i = 1$ if “treated” otherwise 0
- *Sharp RD*: $D_i = 0$ if $x_i < c$; $D_i = 1$ if $x_i \geq c$, 0 otherwise.
 - note: can always normalize x and set $c = 0$
- Potential outcomes $Y_i(0)$ if $D_i = 0$; $Y_i(1)$ if $D_i = 1$
- *Key assumption*:
 - $E[Y_i(0)|x_i]$ is continuous function of x at $x = 0$ (the cutoff)
 - $E[Y_i(1)|x_i]$ is continuous function of x at $x = 0$

RD Framework

- Medicare example: given a cross-sectional data with some people below age $x_i = 65$ and some above 65, you could look at the changes in $Y_i = \text{Hip Replacement}$:



- Define

$$\beta_1 = E[Y_i(1) - Y_i(0) | x_i = 0]$$

This is the treatment effect on agents with $x = 0$ (at the cutoff). Notice that for $x_i < 0$ we have realized $Y_i = Y_i(0)$, and for $x_i \geq 0$ we have $Y_i = Y_i(1)$. So:

- we can estimate $E[Y_i(0)|x_i]$ using data for $x < 0$
- we can estimate $E[Y_i(1)|x_i]$ using data for $x \geq 0$

Then we compare these estimates at $x = 0$ and form $\hat{\beta}_1$.

RD Framework - Example 1

- Consider this special case (linear CEF + constant treatment effect $Y_i(1) - Y_i(0) \equiv \beta_1$):

$$\begin{aligned}E[Y_i(0)|x] &= \alpha_0 + \alpha_1 x \\E[Y_i(1)|x] &= E[Y_i(0)|x] + \beta_1\end{aligned}\tag{1}$$

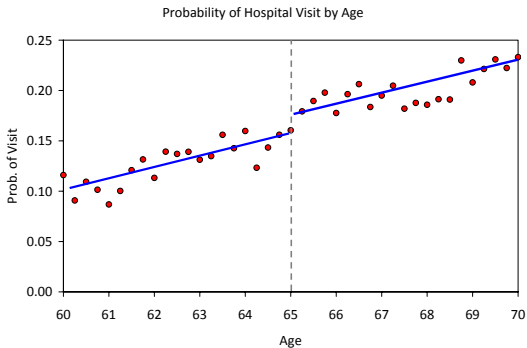
- In this case we can estimate a very simple model:

$$\begin{aligned}Y_i &= \begin{cases} \alpha_0 + \alpha_1 x_i + u_{0i} & \text{if } x_i < 0 \\ \alpha_0 + \alpha_1 x_i + \beta + u_{1i} & \text{if } x_i \geq 0 \end{cases} \\&= \alpha_0 + \alpha_1 x_i + \beta_1 D_i + u_i \text{ for } D_i = 1[x_i \geq 0]\end{aligned}\tag{2}$$

Based on the model (1), we have $E[u_{0i}|x_i] = 0$, $E[u_{1i}|x_i] = 1$, and $u_i = D_i u_{1i} + (1 - D_i) u_{0i}$ is also IND of x_i . What does that imply for our sample OLS?

- Note in this simple model, the slope on x_i is α_1 below and above the cutoff.

RD Framework - Example 1



- Slightly more general model (slope changes at $x = 0$)

$$\begin{aligned} Y_i &= \begin{cases} \alpha_0 + \alpha_1 x_i + u_{0i} & \text{if } x_i < 0 \\ \gamma_0 + \gamma_1 x_i + u_{1i} & \text{if } x_i \geq 0 \end{cases} \\ &= \alpha_0 + \underbrace{(\gamma_0 - \alpha_0)}_{\beta_1} D_i + \alpha_1 x_i + (\gamma_1 - \alpha_1) D_i x_i + u_i \\ &= \beta_0 + \beta_1 D_i + \beta_2 x_i + \beta_3 D_i x_i + u_i \end{aligned} \tag{3}$$

(3) satisfy:

$$\beta_2 = \alpha_1 \text{ slope when } x < 0$$

$$\beta_2 + \beta_3 = \gamma_1 \text{ slope when } x \geq 0$$

- Could also have quadratic x^2 (or cubic x^3) with different coefficients on each side of $x = 0$ cutoff. Note critical assumption that $c = 0$ (e.g., define $x = \text{age} - 65$, the cutoff for Medicare is 0).

Identifying Assumptions

- **Identifying assumption:**

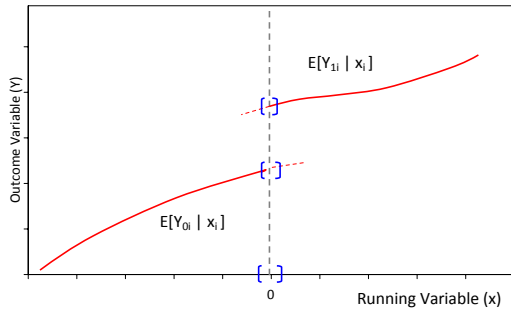
$E[Y_i(0)|x_i]$ and $E[Y_i(1)|x_i]$ are continuous functions of x at “discontinuity” $x = 0$. What does it mean?

- Consider sample 1 with $x_i \in (-\theta, 0)$ and associated $(Y_i(0), Y_i(1))$ vs. sample 2 with $x_i \in [0, \theta)$ and associated $(Y_i(0), Y_i(1))$
 - $\Rightarrow$ distributions of $Y_i(0)$ from 2 samples are centered at $\sim$ same mean
 - and distributions of $Y_i(1)$ from 2 samples are centered at $\sim$ same mean

$\Rightarrow$ *Can use $Y_i(0)$ for sample 1 as “control group” for $Y_i(1)$'s observed in sample 2.*

Identifying Assumptions

Continuity Assumptions of RD Model



Identifying Assumptions

- A sufficient condition for the required continuity is that around $x = 0$, individual assignment to the plus ($x > 0$) or minus ($x < 0$) side is **as good as random**.
- This condition has a **testable implication**:
 - if w_i is some predetermined characteristic (determined prior to assignment), then $E[w_i|x_i]$ will be continuous at $x = 0$.
 - we also expect the relative number of observations to trend smoothly at $x = 0$. No bunching/manipulation around cutoff.

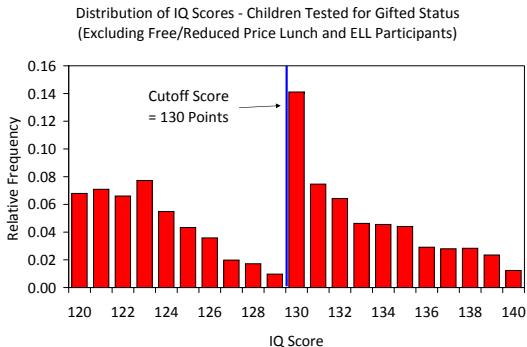
Identifying Assumptions

- How could the RD assumptions break down? Think of the case where $D_i = 1$ is desirable (you win a scholarship if your test score is above a cutoff). If people can bribe the test-taker – or take the test again – we would expect to see:
 - (a) differences in characteristics of people on either side of the cutoff
 - (b) a “spike” of people just to the right

Intuitively, if assignment to the plus or minus side is as good as random, we can think of the people **on either side as treatments and controls in a randomized experiment!**

Identifying Assumptions

- Researchers therefore look for evidence of “manipulation” of the running variable.



Example: Credit Access and College Enrollment

- Alex Solis (2017) asks “Does access to credit explain the gap in schooling attainment between children from richer and poorer families?”. Measuring the effects of credit constraints on college enrollment is challenging:
 - hard to determine whether a family has access to credit;
 - many other **unobservables and self-selection** that affect college enrollment that are likely to be correlated with access to credit -> biased estimates. For example, kids from richer families also have (a) better access to credit, and (b) stronger preferences and preparation for college education.
- He exploits a sharp eligibility rules for college loans in Chile and applies a sharp RD design to estimate the causal impact of **credit access on college enrollment and persistence**.

Example: Credit Access and College Enrollment

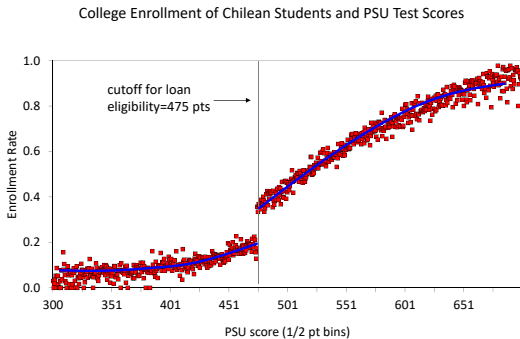
- Setting:
 - in Chile, students take “PSU” test (96% of HS grads take the test)
 - colleges and universities use PSU + GPA to rank students and admit
 - students with family taxable income < 75th percentile can use a loan program if they score **at least 475 points** (combined math and language) on PSU.
 - -sample = 475,000 students who wrote the test 2007-2009.
Administrative data followup to measure enrollment in first year of college/university.
- “Crossing the 475-point PSU test threshold implies a sharp change in access to tuition loans.”

running var: $score_i$

eligibility for loan: $D_i = 1[score_i \geq 475]$

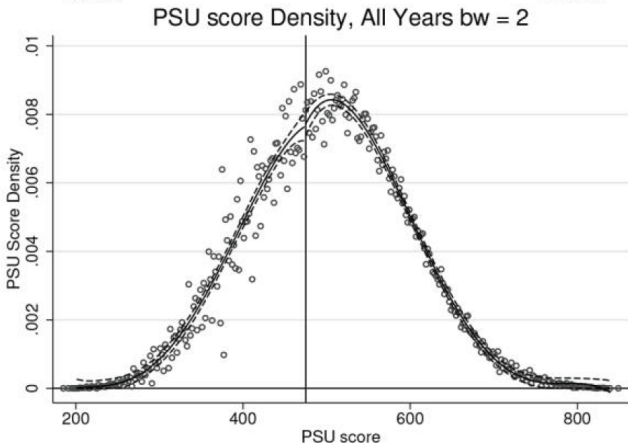
Example: Credit Access and College Enrollment

- Is there a jump in college enrollment?



Example: Credit Access and College Enrollment

- Examine the identifying assumptions for RD:
 - Manipulation of scores around cutoff?



Example: Credit Access and College Enrollment

- Jumps in other characteristics around cutoff?

TABLE
BALANCE OF COVARIATES: POPULATION

VARIABLE	POPULATION				
	Mean (1)	SD (2)	Level (3)	Diff. (4)	SE (5)
Student characteristics:					
High school GPA	56.30	8.54	54.0	-.191	(.100)*
Public school	.45	.50	.44	.005	(.005)
Voucher school	.52	.50	.52	-.005	(.005)
Private school	.03	.17	.04	.000	(.002)
Female	.57	.50	.54	.009	(.005)*
Parent education:					
Mother years of education	11.09	3.51	10.95	-.013	(.038)
Father years of education	11.12	3.73	11.02	-.003	(.042)
Mother dropout high school	.41	.49	.44	-.001	(.005)
Father dropout high school	.44	.50	.45	-.002	(.005)
Mother college graduate	.07	.25	.06	.003	(.003)
Father college graduate	.07	.26	.07	.000	(.003)

Example: Credit Access and College Enrollment

- Concerns about those who take the PSU test multiple times? Those who score below 475 the first time may retake the exam next year and become eligible for student loan in the future
 $\Rightarrow$ eligibility for loan $D_i \neq 1[\text{score}_i \geq 475]$.
- He uses a fuzzy RD:

running var: $z_i = \text{test score on the } \mathbf{1st\ attempt} - 475$

as instrument for eligibility

$D_i = \mathbf{ever}$ being eligible for loans

The ratio (reduced-form over first-stage):

$$\frac{\lim_{z \rightarrow 0^+} Y_i - \lim_{z \rightarrow 0^-} Y_i}{\lim_{z \rightarrow 0^+} D_i - \lim_{z \rightarrow 0^-} D_i}$$

is a local estimate of the impact of credit access on college enrollment, **local to the margin** - score around 475.

Question: What is the denominator if we have a sharp RD?

Example: Credit Access and College Enrollment

- This works as long as not everyone who score just below 475 retake the exam AND are able to score above 475. $\Rightarrow$ “There will still be a discontinuous jump in the ultimate availability of loans at the threshold”. $\lim_{z \rightarrow 0^+} D_i - \lim_{z \rightarrow 0^-} D_i > 0$.
- We will now look at Fuzzy RD in detail.

- In many cases, we don't have a deterministic assignment rule. Instead the probability of assignment to “treatment” depends on the running variable x and jumps discontinuously at $x = 0$. This leads to *Fuzzy RD*:
- *Assignment/First-stage*:

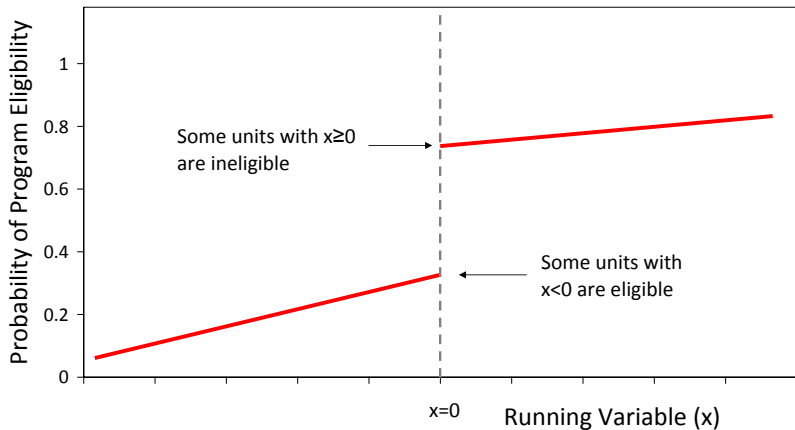
$$Pr(D_i = 1|x_i) = \begin{cases} g_0(x_i) & \text{for } x_i < 0 \\ g_1(x_i) & \text{for } x_i \geq 0 \end{cases}$$

with $g_1(0) \neq g_0(0)$

- Define $z_i = 1[x_i \geq 0]$. (note that $z_i = D_i$ in the 'sharp' case). Then

$$Pr(D_i = 1|x_i) = g_0(x_i) + (g_1(x_i) - g_0(x_i)) z_i$$

Fuzzy RD Based on Discontinuity in Program Eligibility



Identifying Assumptions for Fuzzy RD

- *Key assumptions for fuzzy RD:*
 - ① $E[Y_i(0)|x_i]$ and $E[Y_i(1)|x_i]$ are continuous functions of x at $x = 0$ (threshold).
 - ② First stage: $g_1(0) - g_0(0) \neq 0$, g_1 continuous from above, g_0 continuous from below
- Define:

$$\begin{aligned} \text{(reduced form)} \quad \delta_1 &= \lim_{x \rightarrow 0^+} E[y_i|x_i] - \lim_{x \rightarrow 0^-} E[y_i|x_i] \\ \text{(first stage)} \quad \pi_1 &= \lim_{x \rightarrow 0^+} E[D_i|x_i] - \lim_{x \rightarrow 0^-} E[D_i|x_i] \\ &= g_1(0) - g_0(0) \end{aligned}$$

Then we estimate δ_1 and π_1 and form the ratio δ_1/π_1 . Note that for a sharp RD, $\pi_1 = 1$. Otherwise, $\pi_1 = g_1(0) - g_0(0) < 1$.

Fuzzy RD - Reduced Form

Why does this work? $y_i = Y_i(1) \times D_i + Y_i(0) \times (1 - D_i)$

- From below:

$$\begin{aligned}x_i < 0 : E[y_i|x_i] &= E[Y_i(1)|x_i]g_0(x_i) + E[Y_i(0)|x_i](1 - g_0(x_i)) \\ \Rightarrow \lim_{x \rightarrow 0^-} E[y_i|x_i] &= E[Y_i(1)|x_i = 0]g_0(0) + E[Y_i(0)|x_i = 0](1 - g_0(0))\end{aligned}$$

- From above:

$$\begin{aligned}x_i \geq 0 : E[y_i|x_i] &= E[Y_i(1)|x_i]g_1(x_i) + E[Y_i(0)|x_i](1 - g_1(x_i)) \\ \Rightarrow \lim_{x \rightarrow 0^+} E[y_i|x_i] &= E[Y_i(1)|x_i = 0]g_1(0) + E[Y_i(0)|x_i = 0](1 - g_1(0))\end{aligned}$$

- Reduced form:

$$\begin{aligned}\delta_1 &= (E[Y_i(1)|x_i = 0] - E[Y_i(0)|x_i = 0]) (g_1(0) - g_0(0)) \\ &= E[Y_i(1) - Y_i(0)|x_i = 0] \times \pi_1\end{aligned}$$

Example in Chile revisited

- Solis (2024) estimates a fuzzy RD in which the student's PSU score ($z_i = \text{first score} - 475$) on the **first attempt** serves as an instrument for $D_i = \text{ever being eligible for tuition loan}$. There are students who re-take the exam to get above the threshold, so it's not a sharp RD.
- His two-stage least squares regression:

$$D_i = \pi_0 + \underbrace{\pi_1 1[z_i \geq 0]} + f(z_i) + \eta_i$$

$$Y_i = \beta_0 + \beta_1 D_i + h(z_i) + \nu_i.$$

The control function f in first-stage is a flexible nonlinear function: $f(z_i) = f_+(z_i) \times 1[z_i \geq 0] + f_-(z_i)$. The control function in 2nd stage: $h(z_i) = h_+(z_i) \times 1[z_i \geq 0] + h_-(z_i)$ with continuous h_+, h_- .

- How to estimate f, h ? Bandwidth selection – classic research topic in applied econometrics!